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Section: 5-B

Reg no: 2023-BSE-042

Lab 5

TASK 01

1. Run the java command to see what the system suggests:

java

```
manahil@ubuntu:~$ java
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1, or
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install default-jre             # version 2:1.17-75
sudo apt install openjdk-11-jre-headless # version 11.0.28+6-1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless  # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
manahil@ubuntu:~$ _
```

2. Use the suggested apt command (copy the exact package name suggested by the system). Example (replace with the name shown on your VM):

sudo apt install <suggested-package> -y

```

Adding debian:SSL.com_EV_Root_Certification_Authority_ECC.pem
Adding debian:SSL.com_EV_Root_Certification_Authority_RSA_R2.pem
Adding debian:SSL.com_Root_Certification_Authority_ECC.pem
Adding debian:SSL.com_Root_Certification_Authority_RSA.pem
Adding debian:SSL.com_TLS_ECC_Root_CA_2022.pem
Adding debian:SSL.com_TLS_RSA_Root_CA_2022.pem
Adding debian:Starfield_Class_2_CA.pem
Adding debian:Starfield_Root_Certificate_Authority_-_G2.pem
Adding debian:Starfield_Services_Root_Certificate_Authority_-_G2.pem
Adding debian:SwissSign_Gold_CA_-_G2.pem
Adding debian:SwissSign_Silver_CA_-_G2.pem
Adding debian:SZAFIR_ROOT_CA2.pem
Adding debian:Telia_Root_CA_v2.pem
Adding debian:TeliaSonera_Root_CA_v1.pem
Adding debian:TrustAsia_Global_Root_CA_G3.pem
Adding debian:TrustAsia_Global_Root_CA_G4.pem
Adding debian:Trustwave_Global_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P256_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P384_Certification_Authority.pem
Adding debian:T-TeleSec_GlobalRoot_Class_2.pem
Adding debian:T-TeleSec_GlobalRoot_Class_3.pem
Adding debian:TUBITAK_Kamu_SM_SSL_Kok_Sertifikasi_-_Surum_1.pem
Adding debian:TunTrust_Root_CA.pem
Adding debian:TWCA_Global_Root_CA.pem
Adding debian:TWCA_Root_Certification_Authority.pem
Adding debian:UCA_Extended_Validation_Root.pem
Adding debian:UCA_Global_G2_Root.pem
Adding debian:USERTrust_ECC_Certification_Authority.pem
Adding debian:USERTrust_RSA_Certification_Authority.pem
Adding debian:vTrus_ECC_Root_CA.pem
Adding debian:vTrus_Root_CA.pem
Adding debian:XRamp_Global_CA_Root.pem
done.
Setting up openjdk-21-jre:amd64 (21.0.8+9~us1-0ubuntu1~24.04.1) ...
Setting up default-jre-headless (2:1.21-75+exp1) ...
Setting up default-jre (2:1.21-75+exp1) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
manahil@ubuntu:~$ _

```

3. Verify Java is installed and check version:

`java --version`

```

manahil@ubuntu:~$ java --version
openjdk 21.0.8 2025-07-15
OpenJDK Runtime Environment (build 21.0.8+9-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
manahil@ubuntu:~$

```

4. Remove the Java package using apt remove (use the same package name you installed):

`sudo apt remove <suggested-package> -y`

```

manahil@ubuntu:~$ sudo apt remove default-jre -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dc
  default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hico
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1
  libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolorid2 libcups2t64 libdatatr1 lib
  libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri libglv
  libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 libllvm20 libpango-1.0-0 libpangocairo-1.
  libpixman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-client0 lib
  libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape
  libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxr
  libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-jre-headl
  x11-utils
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  default-jre
0 upgraded, 0 newly installed, 1 to remove and 31 not upgraded.
After this operation, 6,144 B disk space will be freed.
(Reading database ... 139585 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
manahil@ubuntu:~$

```

5. Confirm java is no longer available (run java again) — it should again indicate "not found" or suggest installation:

java

```

manahil@ubuntu:~$ java
-bash: /usr/bin/java: No such file or directory
manahil@ubuntu:~$ _

```

6. Clear the shell's command hash cache so the shell forgets cached command locations (run as your regular user — no sudo required):

TASK 02

1. Install Java using apt-get (choose a common package, e.g., default-jre — or the same package you used in Task 1):

```
sudo apt-get update
```

```

manahil@ubuntu:~$ sudo apt-grt update
sudo: apt-grt: command not found
manahil@ubuntu:~$ sudo apt-get update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,260 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,535 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [292 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.3 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,164 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [488 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,496 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [301 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [208 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.3 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,148 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Ign:15 http://security.ubuntu.com/ubuntu noble-security/main Translation-en
Get:23 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [8,968 B]
Get:25 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,069 kB]
Get:26 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [468 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:28 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [500 B]
Get:29 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [904 kB]
Get:30 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [202 kB]
Get:31 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:32 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.3 kB]
Get:33 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [27.4 kB]
Ign:33 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages
Get:34 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5,708 B]
Get:35 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:36 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [384 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [208 kB]
Get:33 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [27.4 kB]
Fetched 12.3 MB in 13s (933 kB/s)
Reading package lists... Done
manahil@ubuntu:~$ _

```

`sudo apt-get install default-jre -y`

```

0 packages can be upgraded. Run 'apt list --upgradable' to see them.
manahil@ubuntu:~$ sudo apt-get install default-jre -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  default-jre
0 upgraded, 1 newly installed, 0 to remove and 30 not upgraded.
Need to get 922 B of archives.
After this operation, 6,144 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 default-jre amd64 2:1.21-75+exp1 [922 B]
Fetched 922 B in 1s (1,721 B/s)
Selecting previously unselected package default-jre.
(Reading database ... 139570 files and directories currently installed.)
Preparing to unpack .../default-jre_2%3a1.21-75+exp1_amd64.deb ...
Unpacking default-jre (2:1.21-75+exp1) ...
Setting up default-jre (2:1.21-75+exp1) ...
Scanning processes...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-85-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
manahil@ubuntu:~$

```

2. Verify Java version again:

`java --version`

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
manahil@ubuntu:~$ java --version
openjdk 21.0.8 2025-07-15
OpenJDK Runtime Environment (build 21.0.8+9-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
manahil@ubuntu:~$

```

3. Remove Java using apt-get remove:

`sudo apt-get remove default-jre -y`

```

openjdk 64-bit server vm (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
manahil@ubuntu:~$ sudo apt-get remove default-jre -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsetting
  default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-the
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 liba
  libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolord2 libcups2t64 libdatrie1 libdconf1 libdr
  libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri libglvnd0 libglx-me
  libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 libllvm20 libpango-1.0-0 libpangocairo-1.0-0 libpango
  libpixman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-client0 libwayland-curs
  libx11-xcb1 libxau7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm
  libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxrandr2 libxre
  libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-jre-headless session-
  x11-utils
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  default-jre
0 upgraded, 0 newly installed, 1 to remove and 30 not upgraded.
After this operation, 6,144 B disk space will be freed.
(Reading database ... 139571 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
manahil@ubuntu:~$

```

4. Clear the terminal hash cache and confirm java is missing:

`hash -r`

java

TASK 03

1. Update the package index (this downloads the latest lists of available packages):

`sudo apt update`

```
manahil@ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
30 packages can be upgraded. Run 'apt list --upgradable' to see them.
manahil@ubuntu:~$
```

2. Upgrade installed packages (this installs available updates for currently installed packages):

`sudo apt upgrade`

```

fwupd-refresh.service is a disabled or a static unit not running, not starting it.
fwupd.service is a disabled or a static unit not running, not starting it.
Setting up systemd-resolved (255.4-1ubuntu8.11) ...
Setting up snapd (2.71+ubuntu24.04) ...
Installing new version of config file /etc/apparmor.d/usr.lib.snapd.snap-confine.real ...
snapd.failure.service is a disabled or a static unit not running, not starting it.
snapd.gpio-chardev-setup.target is a disabled or a static unit not running, not starting it.
snapd.snap-repair.service is a disabled or a static unit not running, not starting it.
Setting up systemd-sysv (255.4-1ubuntu8.11) ...
Setting up ubuntu-drivers-common (1:0.9.7.6ubuntu3.4) ...
Setting up libnss-systemd:amd64 (255.4-1ubuntu8.11) ...
Setting up libpam-systemd:amd64 (255.4-1ubuntu8.11) ...
Processing triggers for initramfs-tools (0.142ubuntu25.5) ...
update-initramfs: Generating /boot/initrd.img-6.8.0-86-generic
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for rsyslog (8.2312.0-3ubuntu9.1) ...
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for install-info (7.1-3build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
 6.8.0-85-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
systemctl restart multipathd.service open-vm-tools.service polkit.service udisks2.service upower.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
manahil @ session #1: apt[13820], login[967]
manahil @ user manager service: systemd[1388]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
manahil@ubuntu:~$ _

```

- Write a short 3–5 sentence explanation describing the difference between apt update and apt upgrade. Put your text into a small file and capture it as a screenshot (do not upload the text file; provide the screenshot):

nano ~/apt_update_vs_upgrade.md

write 3-5 sentences, save and exit

```
manahil@ubuntu:~$ nano ~/apt_update_vsupgrade.md
```

```

apt update downloads the latest packages lists from the repositories, refreshing the local package index.
it doesnot install or upgrade any software.
apt upgradechecks that updated index for newer versions of the packages already installed and installs them.
without running apt upgrade first, apt upgrade may misses the newest available updates.
inshort, update refreshes the list, and upgrade installs the updates.

```

```
manahil@ubuntu:~$ cat ~/apt_update_vsupgrade.md
apt update downloads the latest packages lists from the repositories, refreshing the local package index.
it doesnot install or upgrade any software.
apt upgradechecks that updated index for newer versions of the packages already installed and installs them.
without running apt upgrade first, apt ugrade may misses the newest available updates.
inshort, update refreshes the list, and upgrade installs the updates.
manahil@ubuntu:~$
```

TASK 04

1. Install VS Code via snap (snap may require sudo):

`sudo snap install --classic code`

```
manahil@ubuntu:~$ cat ~/apt_update_vsupgrade.md
apt update downloads the latest packages lists from the repositories, refreshing the local package index.
it doesnot install or upgrade any software.
apt upgradechecks that updated index for newer versions of the packages already installed and installs them.
without running apt upgrade first, apt ugrade may misses the newest available updates.
inshort, update refreshes the list, and upgrade installs the updates.
manahil@ubuntu:~$ sudo snap install --classic code
[sudo] password for manahil:
2025-10-24T04:11:12Z INFO Waiting for automatic snapd restart...
code 7d842fb8 from Visual Studio Code (vscode) installed
manahil@ubuntu:~$
```

2. Verify snap shows the package is installed:

`snap list code`

```
code 7d842fb8 from Visual Studio Code (vscode) installed
manahil@ubuntu:~$ snap list code
Name Version Rev Tracking Publisher Notes
code 7d842fb8 211 latest/stable vscode+ classic
manahil@ubuntu:~$
```

3. Check the installed application's version. On some systems `code --version` is available; also check `snap info`:

`code --version`

```
code 7d842fb8 211 latest/stable vscode+
manahil@ubuntu:~$ code --version
1.105.1
7d842fb85a0275a4a8e4d7e040d2625abbbf7f084
x64
manahil@ubuntu:~$
```



```

manahil@ubuntu:~$ snap info code
name:      code
summary:   Code editing. Redefined.
publisher: Visual Studio Code (vscode)
store-url: https://snapcraft.io/code
contact:   https://twitter.com/code
license:   unset
description: |
  Visual Studio Code is a new choice of tool that combines the
  simplicity of a code editor with what developers need for the core
  edit-build-debug cycle.
commands:
- code
- code.url-handler
snap-id:    Ht0aUHi7ofh9Fbwh6m7jUN2pAy6kzBiu
tracking:   latest/stable
refresh-date: today at 04:12 UTC
channels:
  latest/stable:    7d842fb8 2025-10-15 (211) 346MB classic
  latest/candidate: ↑
  latest/beta:      ↑
  latest/edge:      ↑
installed:         7d842fb8 (211) 346MB classic
manahil@ubuntu:~$

```

4. If the code binary is not in PATH, show where the snap placed it:

```
ls -l /snap/bin | grep code
```

```

manahil@ubuntu:~$ ls -l /snap/bin | grep code
lrwxrwxrwx 1 root root 13 Oct 24 04:12 code -> /usr/bin/snap
lrwxrwxrwx 1 root root 13 Oct 24 04:12 code.url-handler -> /usr/bin/snap
manahil@ubuntu:~$

```

5. IMPORTANT: Do NOT remove VS Code at the end of this task — keep it installed. It will be launched later in Task 5 and Task 6.

TASK 05

1. Update the server (download package lists and apply upgrades):

```
sudo apt update && sudo apt upgrade -y
```

```

manahil@ubuntu:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,536 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [8,984 B]
Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:12 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,148 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 2,570 kB in 6s (415 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gse
  default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hicolor-ico
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64
  libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolord2 libcups2t64 libdatrie1 libdconf1
  libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri libglvnd0 libg
  libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 libl1vm20 libpango-1.0-0 libpangocairo-1.0-0 lib
  libpixman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-client0 libwayland
  libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 libxc
  libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxrandr2 l
  libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-jre-headless ses
  x11-utils
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
manahil@ubuntu:~$ _

```

2. Install XFCE and XFCE goodies (lightweight desktop):

```
sudo apt install xfce4 xfce4-goodies -y
```

```

aspell-autobuildhash: processing: en [en-variant_0].
aspell-autobuildhash: processing: en [en-variant_1].
aspell-autobuildhash: processing: en [en-variant_2].
aspell-autobuildhash: processing: en [en-w_accents-only].
aspell-autobuildhash: processing: en [en-wo_accents-only].
aspell-autobuildhash: processing: en [en_AU-variant_0].
aspell-autobuildhash: processing: en [en_AU-variant_1].
aspell-autobuildhash: processing: en [en_AU-w_accents-only].
aspell-autobuildhash: processing: en [en_AU-wo_accents-only].
aspell-autobuildhash: processing: en [en_CA-variant_0].
aspell-autobuildhash: processing: en [en_CA-variant_1].
aspell-autobuildhash: processing: en [en_CA-w_accents-only].
aspell-autobuildhash: processing: en [en_CA-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-ise-w_accents-only].
aspell-autobuildhash: processing: en [en_GB-ise-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-ize-w_accents-only].
aspell-autobuildhash: processing: en [en_GB-ize-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-variant_0].
aspell-autobuildhash: processing: en [en_GB-variant_1].
aspell-autobuildhash: processing: en [en_US-w_accents-only].
aspell-autobuildhash: processing: en [en_US-wo_accents-only].
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-85-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
manahil @ session #1: login[967]
manahil @ user manager service: systemd[1388]

```

3. Install and enable XRDP (Remote Desktop Protocol server):

```
sudo apt install xrdp -y
```

```

Unpacking pipewire-module-xrdp (0.2-2) ...
Setting up previously unselected package xorgxrdp.
Preparing to unpack .../xorgxrdp_1%3a0.9.19-1_amd64.deb ...
Unpacking xorgxrdp (1:0.9.19-1) ...
Setting up libpipewire-0.3-modules-xrdp:amd64 (0.2-2) ...
Setting up xorgxrdp (1:0.9.19-1) ...
Setting up libfuse2t64:amd64 (2.9.9-8.1build1) ...
Setting up xrdp (0.9.24-4) ...

Generating 2048 bit rsa key...

ssl_gen_key_xrdp1 ok

saving to /etc/xrdp/rsakeys.ini

Created symlink /etc/systemd/system/multi-user.target.wants/xrdp-sesman.service → /usr/lib/systemd/system/xrdp-sesman.service.
Created symlink /etc/systemd/system/multi-user.target.wants/xrdp.service → /usr/lib/systemd/system/xrdp.service.
Setting up pipewire-module-xrdp (0.2-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
 6.8.0-85-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 manahil @ session #1: login[967]
 manahil @ user manager service: systemd[1388]

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

sudo systemctl enable --now xrdp

```

manahil@ubuntu:~$ sudo systemctl enable --now xrdp
Synchronizing state of xrdp.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable xrdp
manahil@ubuntu:~$ _

```

4. Verify XRDP status:

sudo systemctl status xrdp

```

no VM guests are running outdated hypervisor (qemu) binaries on this host.
manahil@ubuntu:~$ sudo systemctl enable --now xrdp
Synchronizing state of xrdp.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable xrdp
manahil@ubuntu:~$ sudo systemctl status xrdp

• xrdp.service - xrdp daemon
  Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
  Active: active (running) since Fri 2025-10-24 04:40:27 UTC; 1min 48s ago
    Docs: man:xrdp(8)
          man:xrdp.ini(5)
  Main PID: 31749 (xrdp)
    Tasks: 1 (limit: 4548)
  Memory: 872.0K (peak: 1.5M)
    CPU: 121ms
  CGroup: /system.slice/xrdp.service
          └─31749 /usr/sbin/xrdp

Oct 24 04:40:26 ubuntu systemd[1]: Starting xrdp.service - xrdp daemon...
Oct 24 04:40:26 ubuntu xrdp[31747]: [INFO ] address [0.0.0.0] port [3389] mode 1
Oct 24 04:40:26 ubuntu xrdp[31747]: [INFO ] listening to port 3389 on 0.0.0.0
Oct 24 04:40:26 ubuntu xrdp[31747]: [INFO ] xrdp_listen_pp done
Oct 24 04:40:26 ubuntu systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) after start: No such f
Oct 24 04:40:27 ubuntu systemd[1]: Started xrdp.service - xrdp daemon.
Oct 24 04:40:28 ubuntu xrdp[31749]: [INFO ] starting xrdp with pid 31749
Oct 24 04:40:28 ubuntu xrdp[31749]: [INFO ] address [0.0.0.0] port [3389] mode 1
Oct 24 04:40:28 ubuntu xrdp[31749]: [INFO ] listening to port 3389 on 0.0.0.0
Oct 24 04:40:28 ubuntu xrdp[31749]: [INFO ] xrdp_listen_pp done
manahil@ubuntu:~$ _

```

5. Configure XRDp to use XFCE session:

```
echo xfce4-session > ~/.xsession
```

```

manahil@ubuntu:~$ _
manahil@ubuntu:~$ echo xfce4-session > ~/.xsession
manahil@ubuntu:~$ _

```

6. Allow RDP through the firewall (if firewall is in use):

```
sudo ufw allow 3389/tcp
```

```

manahil@ubuntu:~$ sudo ufw allow 3389/tcp
Rules updated
Rules updated (v6)
manahil@ubuntu:~$ _

```

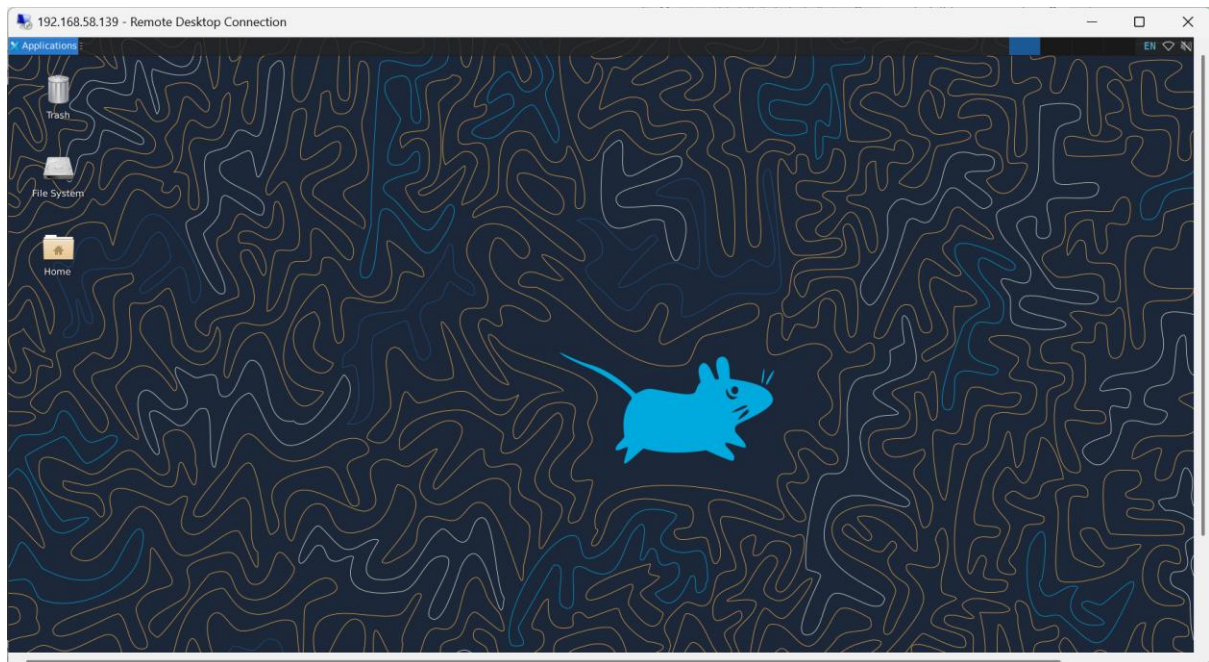
```
sudo ufw reload
```

```

manahil@ubuntu:~$ sudo ufw allow 3389/tcp
Rules updated
Rules updated (v6)
manahil@ubuntu:~$ sudo ufw reload
Firewall not enabled (skipping reload)
manahil@ubuntu:~$

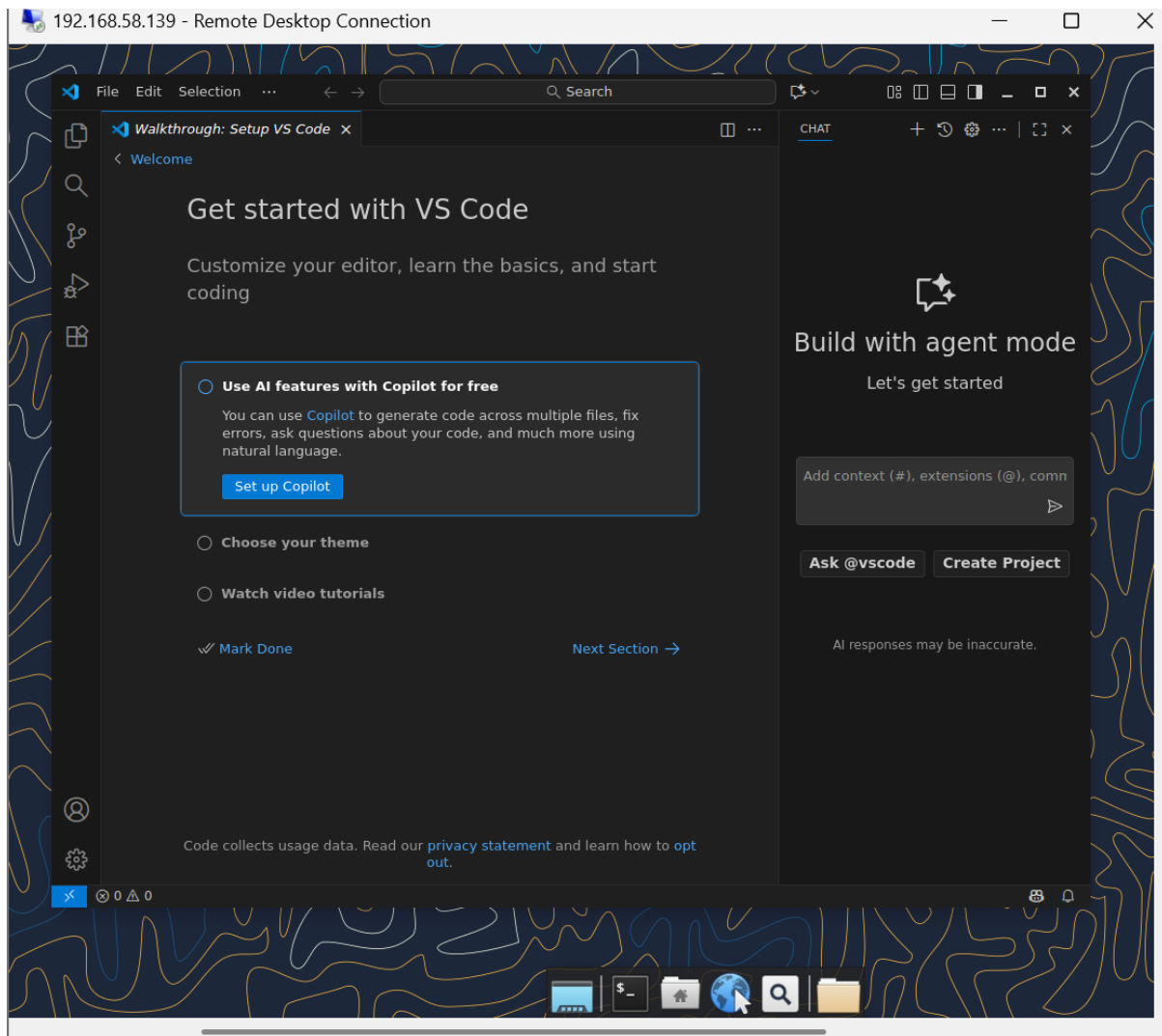
```

7. From a Windows host or RDP client, connect with Remote Desktop (mstsc) to your server IP and login using your Ubuntu username/password. Capture a screenshot of the remote desktop or the RDP session window (if allowed by your environment) and save it as:



8. After you are in the GUI (local console or RDP session), launch Visual Studio Code (installed in Task 4) from the GUI menu or a terminal inside the GUI. Example command from a GUI terminal:

code



- (Optional) From inside VS Code you may open a file or explore the UI — but DO NOT need to create or save files here for Task 5.
9. Fix GUI login screen issues (if lightdm / greeter problems appear)
- Install LightDM and greeter:

```
sudo apt install lightdm lightdm-gtk-greeter -y
```

```

(Reading database ... 176162 files and directories currently installed.)
Preparing to unpack .../gnome-accessibility-themes_3.28-2ubuntu5_all.deb ...
Unpacking gnome-accessibility-themes (3.28-2ubuntu5) ...
Selecting previously unselected package gnome-themes-extra-data.
Preparing to unpack .../gnome-themes-extra-data_3.28-2ubuntu5_all.deb ...
Unpacking gnome-themes-extra-data (3.28-2ubuntu5) ...
Selecting previously unselected package gtk2-engines-pixbuf:amd64.
Preparing to unpack .../gtk2-engines-pixbuf_2.24.33-4ubuntu1.1_amd64.deb ...
Unpacking gtk2-engines-pixbuf:amd64 (2.24.33-4ubuntu1.1) ...
Selecting previously unselected package gnome-themes-extra:amd64.
Preparing to unpack .../gnome-themes-extra_3.28-2ubuntu5_amd64.deb ...
Unpacking gnome-themes-extra:amd64 (3.28-2ubuntu5) ...
Selecting previously unselected package lightdm-gtk-greeter.
Preparing to unpack .../lightdm-gtk-greeter_2.0.9-0ubuntu3_amd64.deb ...
Unpacking lightdm-gtk-greeter (2.0.9-0ubuntu3) ...
Setting up gnome-themes-extra-data (3.28-2ubuntu5) ...
Setting up gnome-accessibility-themes (3.28-2ubuntu5) ...
Setting up gtk2-engines-pixbuf:amd64 (2.24.33-4ubuntu1.1) ...
Setting up lightdm-gtk-greeter (2.0.9-0ubuntu3) ...
update-alternatives: using /usr/share/xgreeters/lightdm-gtk-greeter.desktop to provide /usr/share/xgreeters/light
de
Setting up gnome-themes-extra:amd64 (3.28-2ubuntu5) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-85-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
manahil @ session #1: login[967]
manahil @ user manager service: systemd[1388]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
manahil@ubuntu:~$

```

- Create LightDM config to use XFCE:

```
sudo mkdir -p /etc/lightdm/lightdm.conf.d
```

```
echo -e "[Seat:*]\ngreeter-session=lightdm-gtk-greeter\nuser-
session=xfce\nautologin-user-timeout=0" | sudo tee /etc/lightdm/lightdm.conf.d/99-
xfce.conf
```

```

manahil@ubuntu:~$ echo -e "[Seat:*]\ngreeter-session=lightdm-gtk-greeter\nuser-session=xfce\nautologin-user-timeout=0"
-xfce.conf

[Seat:*]
greeter-session=lightdm-gtk-greeter
user-session=xfce
autologin-user-timeout=0
manahil@ubuntu:~$

```

- Clean up problematic session files and permissions:

```
sudo rm -f /var/lib/lightdm/.Xauthority
```

```
sudo rm -f ~/.Xauthority
```



```
manahil@ubuntu:~$ sudo rm -f /var/lib/lightdm/.Xauthority
manahil@ubuntu:~$ sudo rm -f ~/.Xauthority
manahil@ubuntu:~$ sudo rm -rf ~/.cache/sessions
```

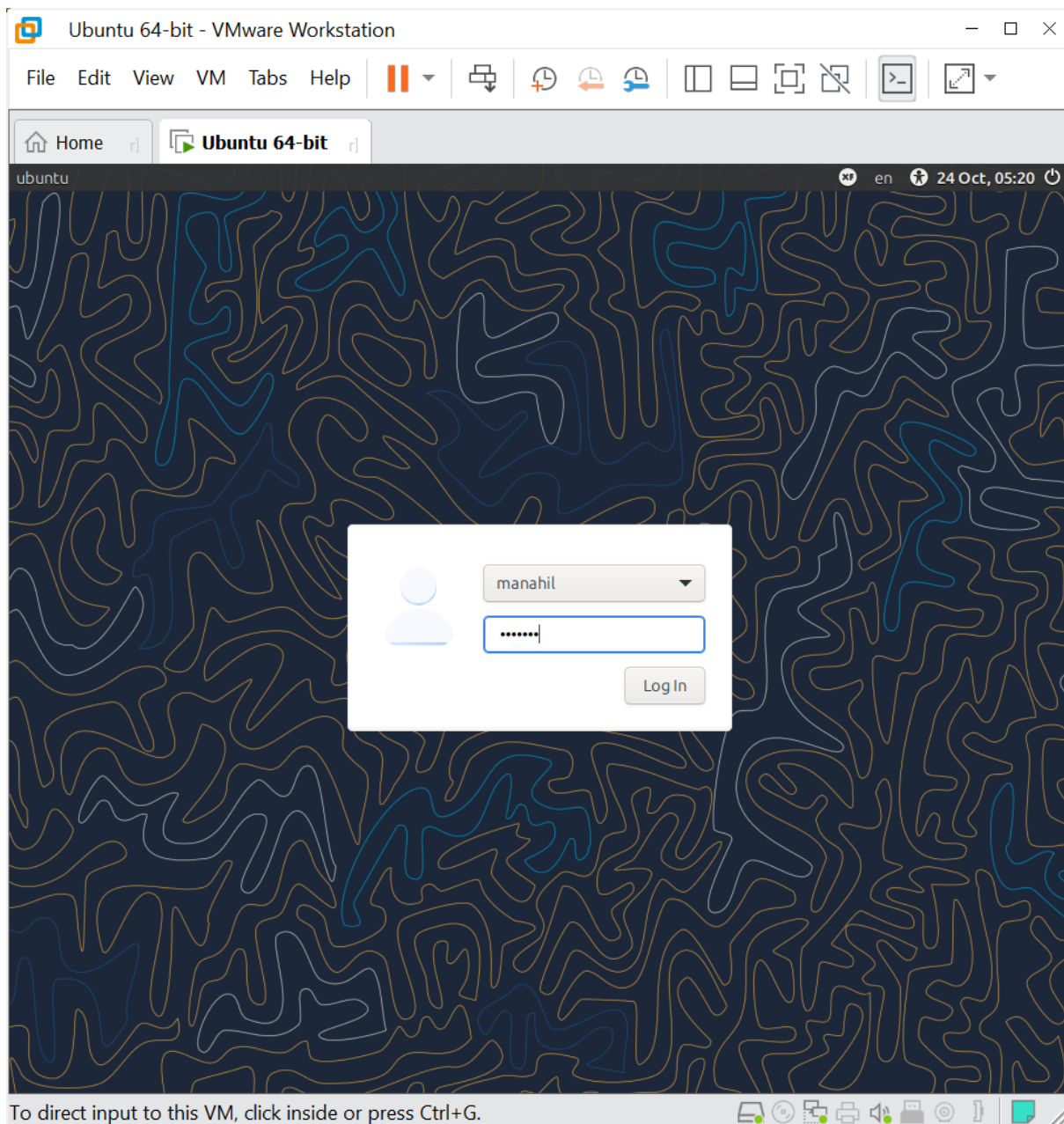
```
sudo rm -rf ~/.cache/sessions
```

```
sudo chown -R $USER:$USER /home/$USER
```

```
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.lockdown.gsch
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.input-sources
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gtk.Settings.FileChooser.gs
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.screensaver.g
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.datetime.gsch
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.a11y.applicat
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.wm.keybinding
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.default-appli
em
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.a11y.keyboard
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.system.locale.gschema
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.enums.xml': R
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gtk.Settings.EmojiChooser.g
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.notifications
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.thumbnails.s
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.wm.preference
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.interface.gsc
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.background.gs
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.search-provid
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.system.location.gsche
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.a11y.mouse.gs
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.app-folders.g
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.thumbnails.g
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.privacy.gsche
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.media-handlin
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.session.gsche
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.a11y.magnifie
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.sound.gschema
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.peripherals.g
chown: changing ownership of '/home/manahil/snap/code/211/.local/share/glib-2.0/schemas/org.gnome.desktop.a11y.gschema.
chown: changing ownership of '/home/manahil/snap/code/common/.cache/gio-modules/libgiognutls.so': Read-only file syst
chown: changing ownership of '/home/manahil/snap/code/common/.cache/gio-modules/libdconfsettings.so': Read-only file sy
chown: changing ownership of '/home/manahil/snap/code/common/.cache/gio-modules/libgiognomeproxy.so': Read-only file sy
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-ti-er.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-vigr.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-am-et.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-broadway.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-cedilla.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-xim.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-cyrillic-translit.so': Read-only file
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-ti-et.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-wayland.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-ipa.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-waylandgtk.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-thai.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-multipress.so': Read-only file system
chown: changing ownership of '/home/manahil/snap/code/common/.cache/immodules/im-inuktitut.so': Read-only file system
chown: cannot access '/home/manahil/thinclient_drives': Permission denied
manahil@ubuntu:~$ _
```

- Restart LightDM:

```
sudo systemctl restart lightdm
```



10. Control GUI login at boot — ENABLE first, then DISABLE (observe and understand terminal/GUI behavior after each reboot)

Important: students MUST perform the reboot after each target change to observe the boot-time behavior. The sequence below has been adjusted so you ENABLE the GUI boot target first, reboot and observe GUI, then DISABLE the GUI boot target, reboot and observe the CLI.

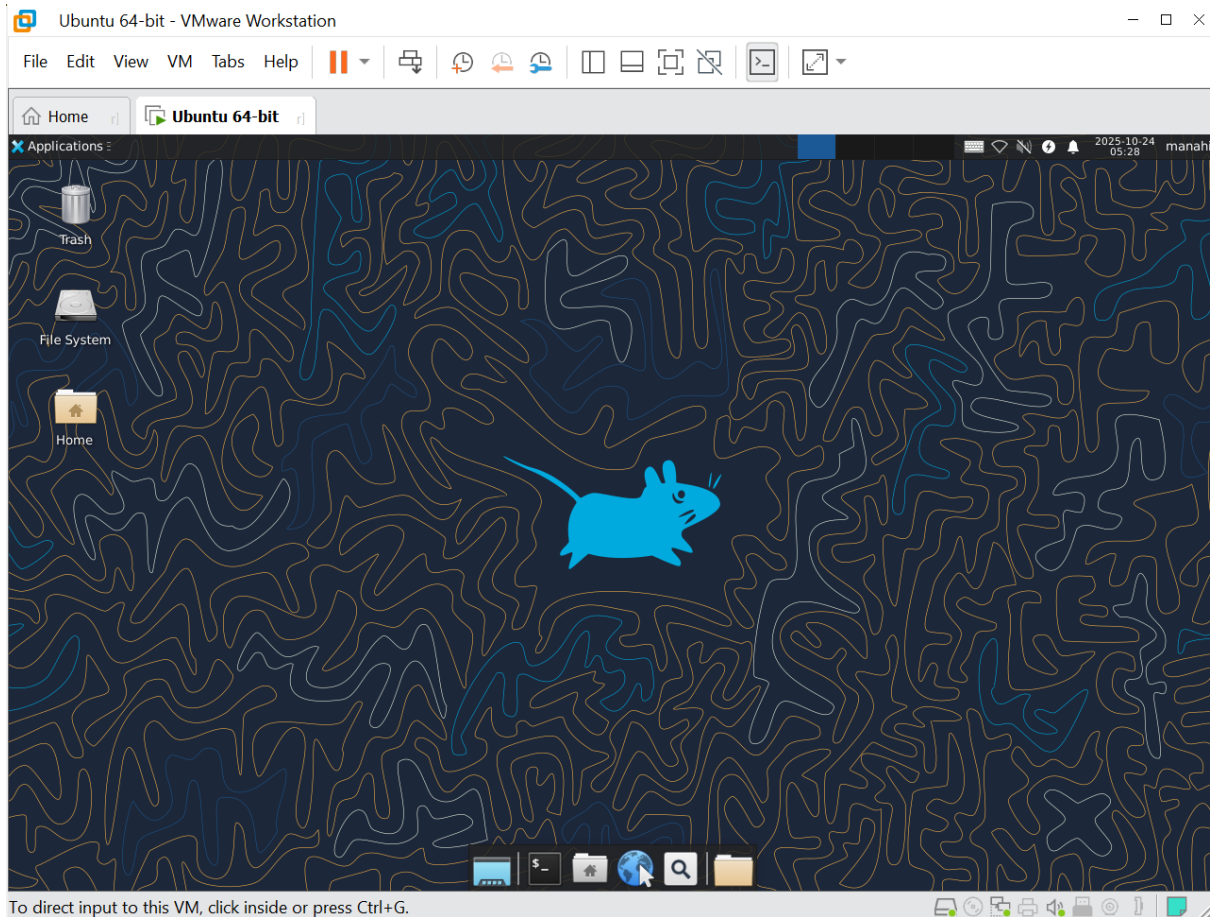
- Enable GUI Login Screen (Boot to GUI)
 - Re-enable LightDM and set the graphical target as default:

```
sudo systemctl enable lightdm
```

```
sudo systemctl set-default graphical.target
```

- Reboot the VM to observe that it boots to the GUI login screen:

sudo reboot



- Disable GUI Login Screen (Boot to CLI)
 - Set the default boot target to multi-user (text mode) and disable LightDM so the system boots to the terminal:

sudo systemctl set-default multi-user.target

sudo systemctl disable lightdm

```
manahil@ubuntu:~$ sudo systemctl set-default multi-user.target
[sudo] password for manahil:
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
manahil@ubuntu:~$ sudo systemctl disable lightdm
sudo: systemctl: command not found
manahil@ubuntu:~$ sudo systemctl disable lightdm
sudo: systemctl: command not found
manahil@ubuntu:~$ sudo systemctl disable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable lightdm
Removed "/etc/systemd/system/display-manager.service".
manahil@ubuntu:~$
```

- Reboot the VM to observe that it boots to the terminal (CLI):

sudo reboot

```
Ubuntu 24.04.3 LTS ubuntu tty1
ubuntu login: manahil
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Fri Oct 24 05:30:27 AM UTC 2025

System load:  0.84               Processes:            314
Usage of /:   70.3% of 9.75GB    Users logged in:     1
Memory usage: 17%               IPv4 address for ens33: 192.168.58.139
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

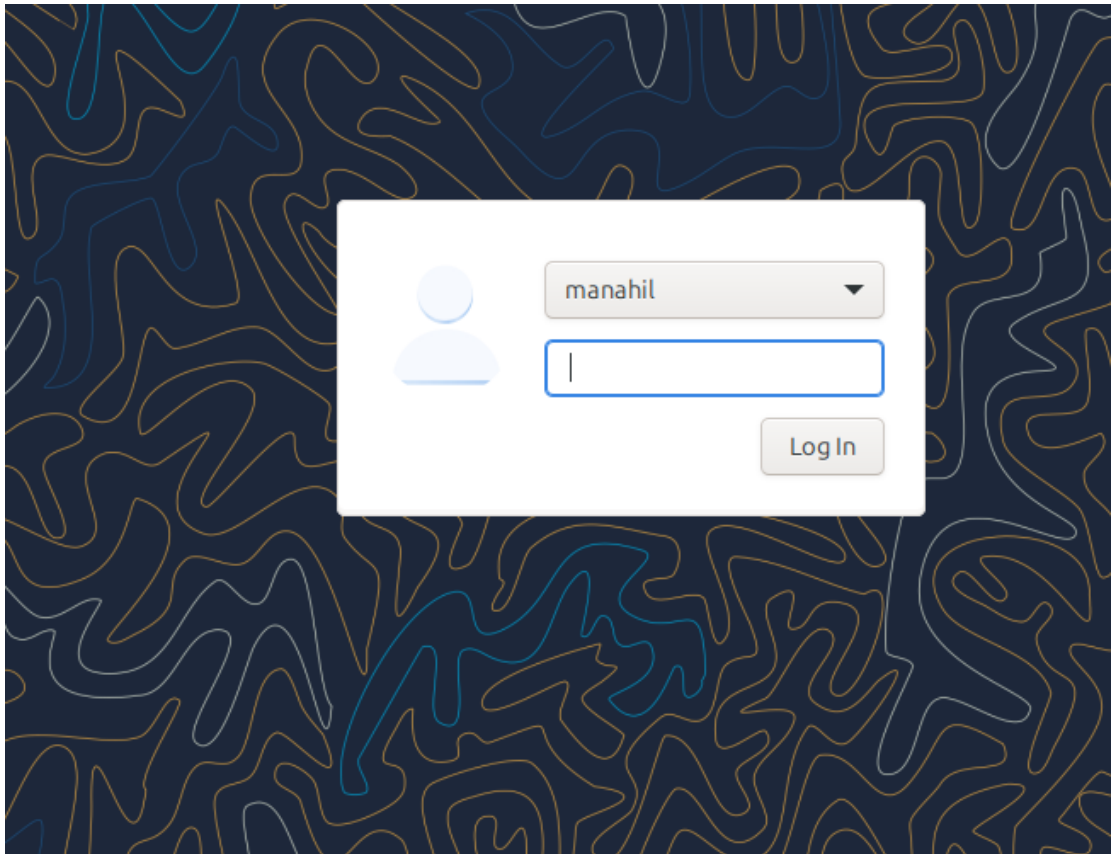
0 updates can be applied immediately.

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

manahil@ubuntu:~$
```

- After the VM boots, capture a screenshot of the login prompt or terminal session showing CLI-only behavior and save it as: task5_after_reboot_cli.png
- Start/Stop GUI manually (no reboot)
 - You can start the GUI session without changing the boot target. This is useful if you want to keep the boot target as CLI but run GUI temporarily:

sudo systemctl start lightdm # start GUI now



- Press Ctrl + Alt + F3 to switch back to TTY. You can stop the GUI session without changing the boot target.

```
sudo systemctl stop lightdm # stop GUI now
```

```
ubuntu login: manahil
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Oct 24 05:42:31 AM UTC 2025

System load:  0.13               Processes:            273
Usage of /:   70.4% of 9.75GB    Users logged in:     1
Memory usage: 10%               IPv4 address for ens33: 192.168.58.139
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

manahil@ubuntu:~$ sudo systemctl stop lightdm

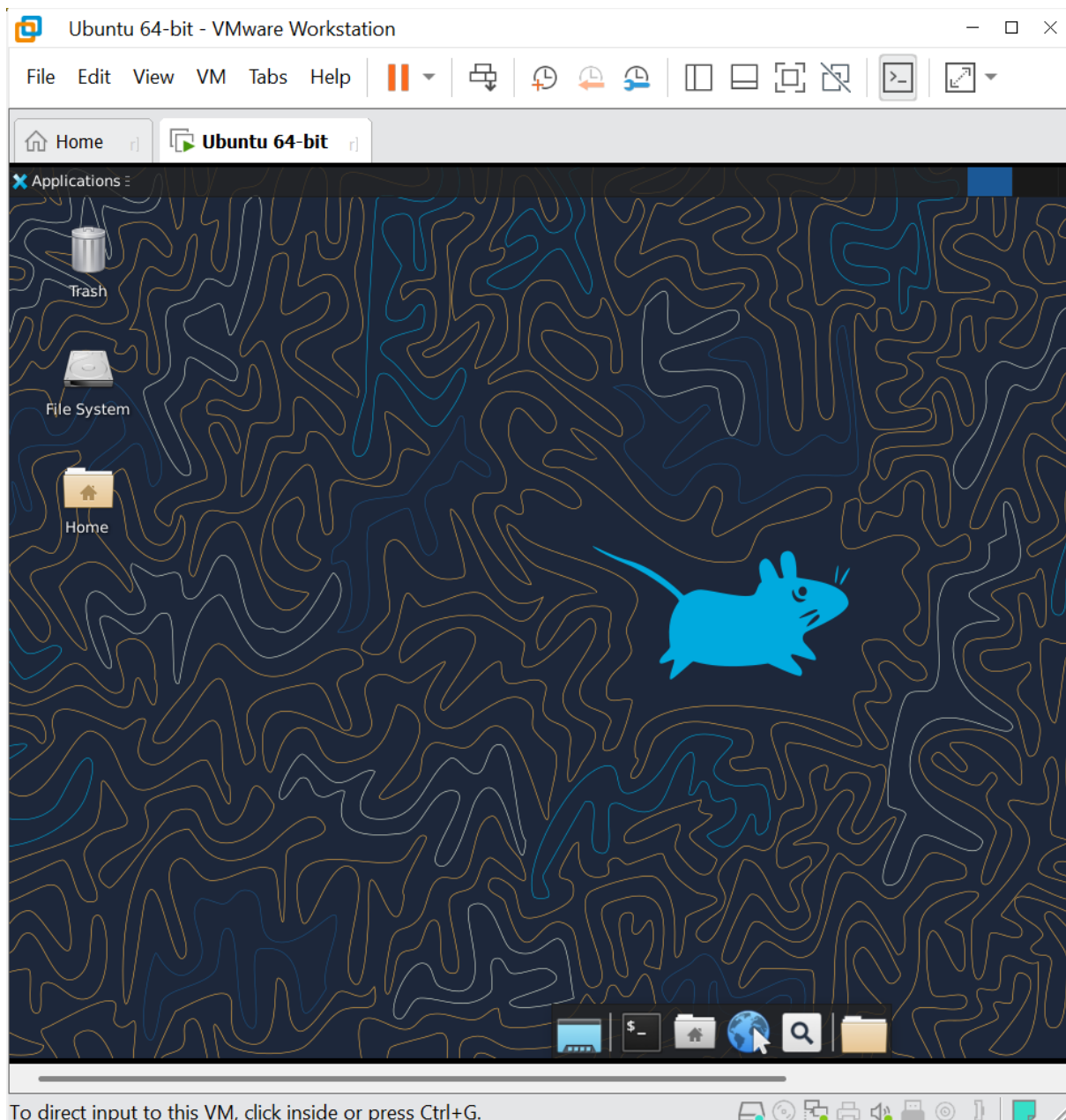
[sudo] password for manahil:
Sorry, try again.
[sudo] password for manahil:
manahil@ubuntu:~$
```

TASK 06

Steps (inside the VM terminal / via SSH)

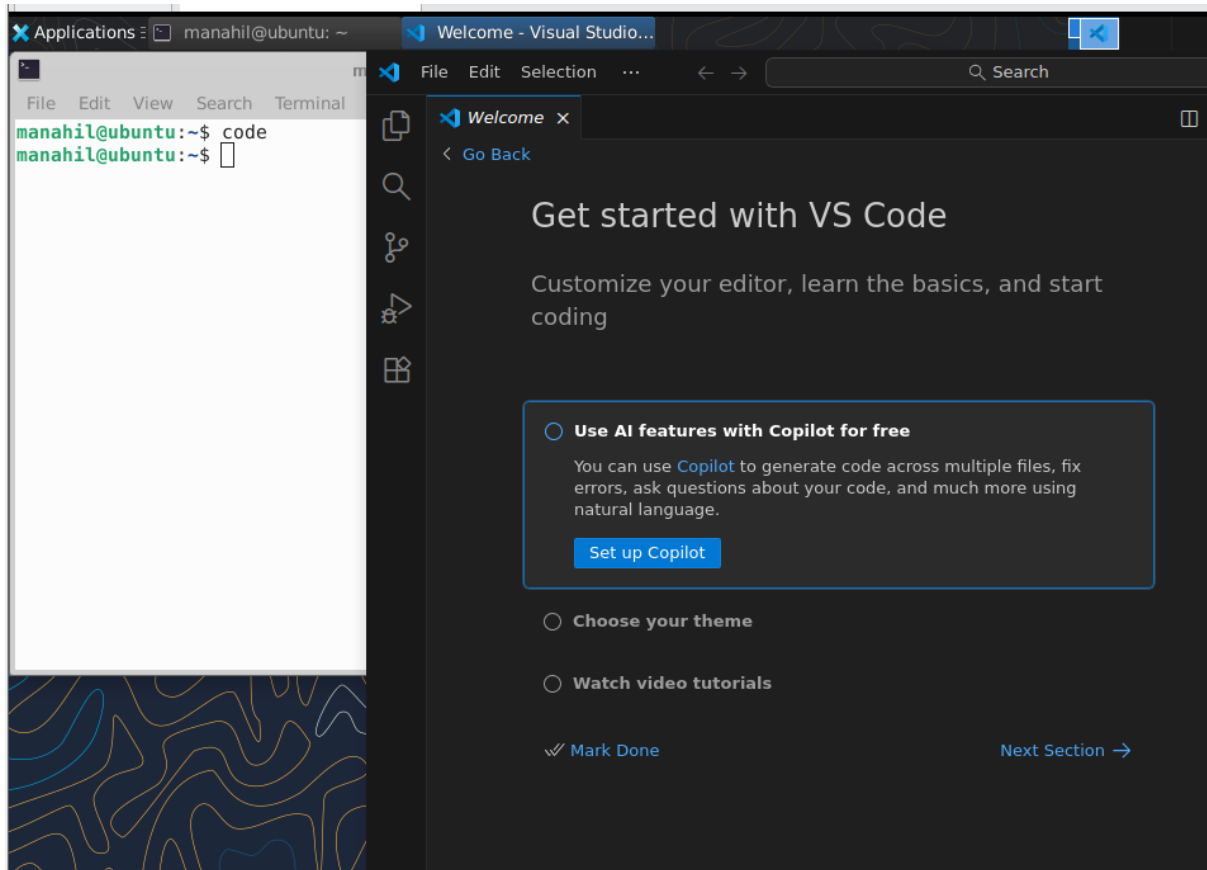
1. Start the GUI (if the system is currently set to CLI or the GUI is not running):

```
sudo systemctl start lightdm
```

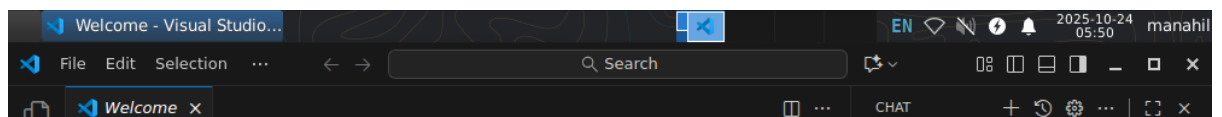


2. In the GUI session (local console or RDP) launch Visual Studio Code (installed earlier in Task 4). From a GUI terminal inside the desktop, run:

code



3. Close VS Code using the GUI window controls. Task 6 ends here.



TASK 07

1. (Learning step — first command must be the install attempt) Attempt to install Google Chrome directly to see the failure when the repo/key are missing:

`sudo apt install google-chrome-stable -y`

- Expected: this will typically fail with "Unable to locate package google-chrome-stable" or similar because the Google repo/key are not yet configured.


```
manahil@ubuntu:~$ code
manahil@ubuntu:~$ code
manahil@ubuntu:~$ sudo apt install google-chrome-stable -y

[sudo] password for manahil:
Sorry, try again.
[sudo] password for manahil:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package google-chrome-stable
manahil@ubuntu:~$
```

2. Inspect apt configuration so you understand why install failed. List the /etc/apt directory:

ls -la /etc/apt

```
manahil@ubuntu:~$ ls -la /etc/apt

total 48
drwxr-xr-x  9 root root  4096 Sep 26 12:17 .
drwxr-xr-x 142 root root 12288 Oct 24 04:55 ..
drwxr-xr-x  2 root root  4096 Oct 24 00:08 apt.conf.d
drwxr-xr-x  2 root root  4096 Mar 31 2024 auth.conf.d
drwxr-xr-x  2 root root  4096 Mar 31 2024 keyrings
drwxr-xr-x  2 root root  4096 Aug  5 17:14 preferences.d
drwxr-xr-x  2 root root  4096 Aug  5 17:14 preferences.d.save
-rw-r--r--  1 root root    70 Sep 26 07:22 sources.list
drwxr-xr-x  2 root root  4096 Sep 26 07:22 sources.list.d
drwxr-xr-x  2 root root  4096 Aug  5 17:01 trusted.gpg.d
manahil@ubuntu:~$
manahil@ubuntu:~$
```

3. View the main /etc/apt/sources.list:

cat /etc/apt/sources.list

```
manahil@ubuntu:~$ cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
manahil@ubuntu:~$
```

4. List files under /etc/apt/sources.list.d:

ls -la /etc/apt/sources.list.d/

```
manahil@ubuntu:~$ ls -la /etc/apt/sources.list.d/

total 16
drwxr-xr-x 2 root root 4096 Sep 26 07:22 .
drwxr-xr-x 9 root root 4096 Sep 26 12:17 ..
-rw-r--r-- 1 root root  383 Sep 26 07:22 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
manahil@ubuntu:~$
manahil@ubuntu:~$
```

5. If there is a file named `ubuntu.sources` (or similarly named source file), display it to see whether Chrome's repo is present:

```
cat /etc/apt/sources.list.d/ubuntu.sources
```

```
manahil@ubuntu:~$ cat /etc/apt/sources.list.d/ubuntu.sources

Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
manahil@ubuntu:~$
```

6. Add Chrome repository metadata to a sources file (method A — using `ubuntu.sources`). Open or create the file and append the stanza (you can alternatively use the preferred one-line method in step 11):

```
sudo nano /etc/apt/sources.list.d/ubuntu.sources
```

```
File Edit View Search Terminal Help
GNU nano 7.2 /etc/apt/sources.list.d/ubuntu.sources *
Types: deb URIs: http://archive.ubuntu.com/ubuntu/ Suites: noble noble-updates noble-backports Components
multiverse Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
Types: deb
URIs: http://dl.google.com/linux/chrome/deb/

Suites: stable

Components: main

Architectures: amd64

Signed-By: /etc/apt/keyrings/google.gpg
█
```

7. Ensure the `keyrings` directory exists and import Google's signing key:

```
curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o
/etc/apt/keyrings/google.gpg
```

```
manahil@ubuntu:~$ sudo mkdir -p /etc/apt/keyrings
manahil@ubuntu:~$ curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /et
manahil@ubuntu:~$
```

8. Update apt and attempt to install Chrome again (now that repo + key are added):

sudo apt update

```
manahil@ubuntu:~$ sudo apt update
Get:1 http://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Get:2 http://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,214 B]
Fetched 3,039 B in 2s (1,436 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

sudo apt install google-chrome-stable -y

9. Alternate (preferred, cleaner) method — create a single google-chrome.list entry
10. (Optional cleanup before alternate method) If you added the chrome stanza earlier and want to switch to the preferred method:

sudo apt remove google-chrome-stable -y

```
manahil@ubuntu:~$ sudo apt remove google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  google-chrome-stable
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
1 not fully installed or removed.
After this operation, 396 MB disk space will be freed.
(Reading database ... 180111 files and directories currently installed.)
Removing google-chrome-stable (141.0.7390.122-1) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
needrestart is being skipped since dpkg has failed
manahil@ubuntu:~$ sudo nano /etc/apt/sources.list.d/ubuntu.sources
```

sudo nano /etc/apt/sources.list.d/ubuntu.sources # remove the chrome stanza you added earlier, save and exit

sudo rm -f /etc/apt/keyrings/google.gpg

- Save screenshots as: task7_alternate_remove.png, task7_alternate_edit.png, and task7_remove_key.png

11. Create a dedicated one-line list file for Google Chrome (preferred):

```
echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/google.gpg]
http://dl.google.com/linux/chrome/deb/ stable main" | sudo tee
/etc/apt/sources.list.d/google-chrome.list > /dev/null
```

- Save screenshot as: task7_create_google_chrome_list.png

12. Verify the new file exists:

ls -la /etc/apt/sources.list.d/

- Save screenshot as: task7_list_sources_after_create.png

13. Re-add the Google signing key (if removed previously or not present):

```
sudo mkdir -p /etc/apt/keyrings
```

```
curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
```

- Save screenshot as: task7_add_key_alt.png

14. Update apt and install Chrome (preferred flow):

```
sudo apt update
```

```
sudo apt install google-chrome-stable -y
```

- Save screenshots as: task7_apt_update_alt.png and task7_install_chrome_alt.png

 Screenshot Required:

TASK 08

1. Add the Audacity PPA, update apt and install audacity:

```
sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y
```

```
manahil@ubuntu:~$ sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu/
Suites: noble
Components: main
'
Description:
Unofficial build of Audacity audio editor

For help, please use Audacity forum: http://forum.audacityteam.org/

If the packages here are helpful, you may buy me a coffee:

https://ko-fi.com/ubuntuhandbook1
More info: https://launchpad.net/~ubuntuhandbook1/+archive/ubuntu/audacity
Adding repository.
Get:1 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease [18.1 kB]
Hit:2 http://dl.google.com/linux/chrome/deb stable InRelease
Get:3 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main amd64 Packages [1,064 B]
Get:4 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main Translation-en [492 B]
Fetched 19.6 kB in 1s (13.3 kB/s)
Reading package lists... Done
manahil@ubuntu:~$
```

```
sudo apt update
```

```

manahil@ubuntu:~$ sudo apt update
Hit:1 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Hit:2 http://dl.google.com/linux/chrome/deb stable InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
manahil@ubuntu:~$

```

sudo apt install audacity -y

```

manahil@ubuntu:~$ sudo apt install audacity -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Some packages could not be installed. This may mean that you have
requested an impossible situation or if you are using the unstable
distribution that some required packages have not yet been created
or been moved out of Incoming.
The following information may help to resolve the situation:

The following packages have unmet dependencies:
 audacity : Depends: libflac++10 (>= 1.3.1) but it is not installable
            Depends: libid3tag0 (>= 0.15.1b) but it is not installable
            Depends: liblilv-0-0 (>= 0.24.0~dfsg0) but it is not installable
            Depends: libopusfile0 (>= 0.5) but it is not installable
            Depends: libportaudio2 (>= 19+svn20101113) but it is not installable
            Depends: libportmidi0 but it is not installable
            Depends: libportsmf0t64 (>= 0.1~svn20101010) but it is not installable
            Depends: libbsbms10 (>= 2.3.0) but it is not installable
            Depends: libsoundtouch1 (>= 2.0.0) but it is not installable
            Depends: libsuil-0-0 (>= 0.8.0~dfsg0) but it is not installable
            Depends: libvamp-hostsdk3v5 (>= 2.10.0) but it is not installable
            Depends: libwxbase3.2-1t64 (>= 3.2.4+dfsg) but it is not installable
            Depends: libwxgtk3.2-1t64 (>= 3.2.4+dfsg) but it is not installable
E: Unable to correct problems, you have held broken packages.
manahil@ubuntu:~$

```

2. Launch Audacity (from GUI or CLI). On a headless server you may not get a GUI window — if you are using the XFCE GUI session, launch from a GUI terminal or run check for binary:

audacity --version

```

manahil@ubuntu:~$ audacity --version
manahil@ubuntu:~$ audacity --version
manahil@ubuntu:~$ which audacity

/usr/bin/audacity
manahil@ubuntu:~$
manahil@ubuntu:~$

```

3. Add the OBS Studio PPA, update apt and install obs-studio:

sudo add-apt-repository ppa:obsproject/obs-studio -y

```
manahil@ubuntu:~$ sudo add-apt-repository ppa:obsproject/obs-studio -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu/
Suites: noble
Components: main
'
Description:
Latest stable release of OBS Studio
More info: https://launchpad.net/~obsproject/+archive/ubuntu/obs-studio
Adding repository.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://dl.google.com/linux/chrome/deb stable InRelease
Get:6 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease [17.8 kB]
Hit:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Get:8 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main amd64 Packages [1,172 B]
Get:9 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main Translation-en [160 B]
Fetched 19.1 kB in 4s (5,130 B/s)
Reading package lists... Done
manahil@ubuntu:~$
```

sudo apt update

sudo apt install obs-studio -y

```
manahil@ubuntu:~$ sudo apt install obs-studio -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
 libaacs0 libass9 libavdevice60 libavfilter9 libavformat60 libb2-1 libbdplus0 libblas3 libbluray2 li
 libdc1394-25 libdecor-0-0 libdecor-0-plugin-1-gtk libfdk-aac2 libfftw3-double3 libflite1 libgfortra
 liblua5.1-common libmbedcrypto7t64 libmbdts14t64 libmbx509-1t64 libmysofa1 libnorm1t64 libo
 libpgm-5.3-0t64 libplacebo338 libpocketsphinx3 libpostproc57 libqrencode4 libqt6core6t64 libqt6
 libqt6network6t64 libqt6opengl6t64 libqt6qml6 libqt6qmlmodels6 libqt6quick6 libqt6svg6 libqt6waylan
 libqt6waylandeglclienthintegration6 libqt6waylandeglcompositorhintegration6 libqt6widgets6t64 lib
 librabbitmq4 librist4 librubberband2 libSDL2-2.0-0 libsndio7.0 libspnbase3t64 libstr1.5-gnutls l
 libswscale7 libts0t64 libudfread0 libunibreak5 libvidstab1.1 libxcb-composite0 libzim2 libzm5 poc
 qt6-gpa-plugins qt6-translations-l10n qt6-wayland
Suggested packages:
 libbluray-bdj libfftw3-bin libfftw3-dev qt6-qmltooling-plugins sndiod
The following NEW packages will be installed:
 libaacs0 libass9 libavdevice60 libavfilter9 libavformat60 libb2-1 libbdplus0 libblas3 libbluray2 li
 libdc1394-25 libdecor-0-0 libdecor-0-plugin-1-gtk libfdk-aac2 libfftw3-double3 libflite1 libgfortra
 liblua5.1-common libmbedcrypto7t64 libmbdts14t64 libmbx509-1t64 libmysofa1 libnorm1t64 libo
 libpgm-5.3-0t64 libplacebo338 libpocketsphinx3 libpostproc57 libqrencode4 libqt6core6t64 libqt6
 libqt6network6t64 libqt6opengl6t64 libqt6qml6 libqt6qmlmodels6 libqt6quick6 libqt6svg6 libqt6waylan
 libqt6waylandeglclienthintegration6 libqt6waylandeglcompositorhintegration6 libqt6widgets6t64 lib
 librabbitmq4 librist4 librubberband2 libSDL2-2.0-0 libsndio7.0 libspnbase3t64 libstr1.5-gnutls l
 libswscale7 libts0t64 libudfread0 libunibreak5 libvidstab1.1 libxcb-composite0 libzim2 libzm5 obs
 qt6-gtk-platformtheme qt6-gpa-plugins qt6-translations-l10n qt6-wayland
0 upgraded, 75 newly installed, 0 to remove and 0 not upgraded.
Need to get 182 MB of archives.
After this operation, 534 MB of additional disk space will be used.
```

```
manahil@ubuntu:~$ obs --version
OBS Studio - 32.0.0
manahil@ubuntu:~$
```

TASK 09

1. Check whether vim is installed by running:

Vim



The screenshot shows the Vim editor's startup screen. At the top is a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The main area displays the following text:

```
VIM - Vi IMproved
        version 9.1.697
        by Bram Moolenaar et al.
    Modified by team+vim@tracker.debian.org
    Vim is open source and freely distributable

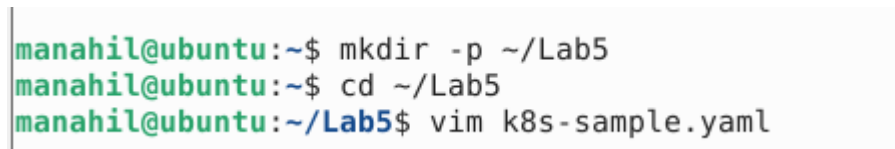
    Sponsor Vim development!
type  :help sponsor<Enter>    for information

type  :q<Enter>                to exit
type  :help<Enter> or <F1>    for on-line help
type  :help version9<Enter>  for version info
```

2. Create the Lab5 working directory in your home and change into it:

```
mkdir -p ~/Lab5
```

```
cd ~/Lab5
```



The terminal screenshot shows the following commands and their output:

```
manahil@ubuntu:~$ mkdir -p ~/Lab5
manahil@ubuntu:~$ cd ~/Lab5
manahil@ubuntu:~/Lab5$ vim k8s-sample.yaml
```

3. Create the Kubernetes sample file using vim:

```
vim k8s-sample.yaml
```

```
manahil@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1

kind: Pod

metadata:
  name: nginx-pod

spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always

manahil@ubuntu:~/Lab5$
```

TASK 10

1. Open the manifest with vim:

cd ~/Lab5

```
manahil@ubuntu:~/Lab5$ cd ~/Lab5
```

vim k8s-sample.yaml

2. Add the annotation under the metadata section (indentation must match YAML).
Example (insert these lines under metadata:):


```

manahil@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1

kind: Pod

metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always

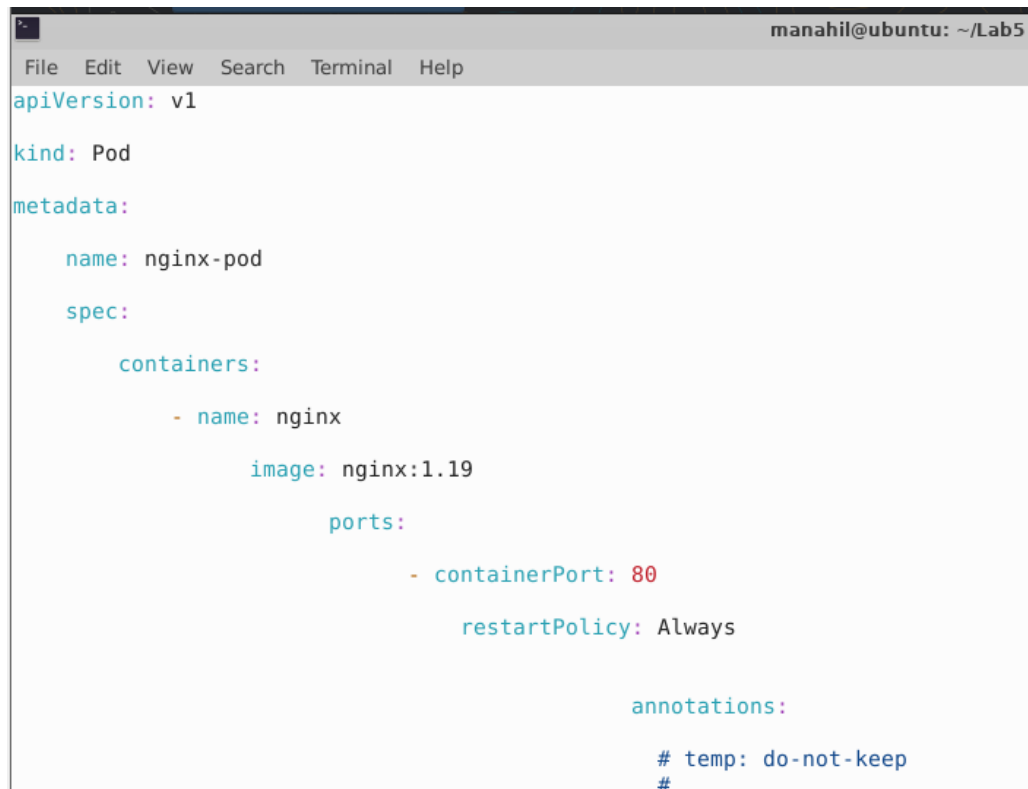
    annotations:
      lab: lesson11
manahil@ubuntu:~/Lab5$ █

```

3. Discard changes (practice: make a temporary edit and exit without saving)

- Open the file again:

vim k8s-sample.yaml



```

manahil@ubuntu: ~/Lab5
File Edit View Search Terminal Help
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always

    annotations:
      # temp: do-not-keep
      #

```

- Verify the file does NOT contain the temporary comment:

cat k8s-sample.yaml

```
manahil@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1

kind: Pod

metadata:
  name: nginx-pod

  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always

    annotations:
      lab: lesson1

manahil@ubuntu:~/Lab5$
```

TASK 11

1. Open the file with vim:

cd ~/Lab5

vim k8s-sample.yaml

2. Delete the line containing image: nginx:1.19 with a single command:

Dd

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        ports:
```

U

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
```

3. Delete 3 lines at once using the numeric delete command:

d3d

```
File Edit View Search Terminal Help
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort:
            restartPolicy
```

U

```
File Edit View Search Terminal Help
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
```

4. Navigation practice (from command mode)

- Jump to the first line:

1G

```
apiVersion: v1
kind: Pod
```

- Jump to the last line:

G

```
annotations:
  lab: lesson11
```

TASK 12

1. Open the file with vim:

```
cd ~/Lab5
```

```
vim k8s-sample.yaml
```

2. Search for the string nginx using the forward search command:

- From command mode type:

```
/nginx
```

```

metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
        annotations:
          lab: lesson11
      -name nginx
~
~
~
~
~
~
~
~
~
~
search hit BOTTOM, continuing at TOP

```

3. Move to the next match and previous match:

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always

```

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx

```

4. Add two more occurrences of the word nginx in the file:

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
        annotations:
          label: nginx
      - name: #nginx
      - name: #nginx

```

~/nginx

5. Demonstrate that n cycles forward through all matches:
6. Substitute all occurrences of nginx with webapp:

```

apiVersion: v1
kind: Pod
metadata:
  name: webapp-pod
  spec:
    containers:
      - name: webapp
        image: webapp:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
    annotations:
      lab: lesson11
      #webapp
      #webapp
~
~
~
~
~
~
~
~
5 substitutions on 5 lines

```

7. Immediately undo the substitution using u:

```

apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
    annotations:
      lab: lesson11
~

```

8. Quit vim without saving any accidental changes (if you didn't already save after substitution and want to discard):

- If you want to ensure no changes were written, exit with:

:q!

```
manahil@ubuntu:~/Lab5$  
manahil@ubuntu:~/Lab5$ vim ~/Lab5/k8s-sample.yaml  
manahil@ubuntu:~/Lab5$ vim ~/Lab5/k8s-sample.yaml  
manahil@ubuntu:~/Lab5$ █
```

Evaluation Task :

```
manahil@ubuntu:~$ docker run hello-world  
  
Hello from Docker!  
This message shows that your installation appears to be working correctly.  
  
To generate this message, Docker took the following steps:  
1. The Docker client contacted the Docker daemon.  
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.  
   (amd64)  
3. The Docker daemon created a new container from that image which runs the  
   executable that produces the output you are currently reading.  
4. The Docker daemon streamed that output to the Docker client, which sent it  
   to your terminal.  
  
To try something more ambitious, you can run an Ubuntu container with:  
$ docker run -it ubuntu bash  
  
Share images, automate workflows, and more with a free Docker ID:  
https://hub.docker.com/  
  
For more examples and ideas, visit:  
https://docs.docker.com/get-started/  
  
manahil@ubuntu:~$ docker --version  
Docker version 28.5.1, build e180ab8  
manahil@ubuntu:~$ █
```
